

Instructor: Date:

Learner: Regional Branch:

Program: Center:

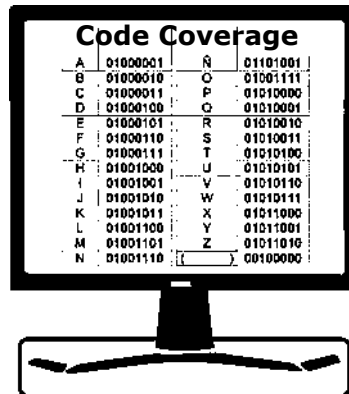
CONTENT

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Let's Start!

We must know what code coverage is used for.



A	01000001	N	01101001
B	01000010	O	01001111
C	01000011	P	01010000
D	01000100	Q	01010001
E	01000101	R	01010010
F	01000110	S	01010011
G	01000111	T	01010100
H	01001000	U	01010101
I	01001001	V	01010110
J	01001010	W	01010111
K	01001011	X	01011000
L	01001100	Y	01011001
M	01001101	Z	01011010
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Learning Outcome:

Describe ways of solving problems and getting things done in written and spoken situations and in work and social contexts.

Train the system users on application structuring and management, according to the established plan and user profile, according to the organization policies.

Ask your partner:

- When do you think it is not necessary to test code?
- Why do you think it is important to know exactly what metrics measure? Why should we be critical with the metrics we use?

Can Do Statements:

- I can talk about the usefulness of code coverage as a measure.
- I can discuss how the way I test is a critical component along with what I test.
- I can talk about exercising caution in how I interpret the results with code coverage.



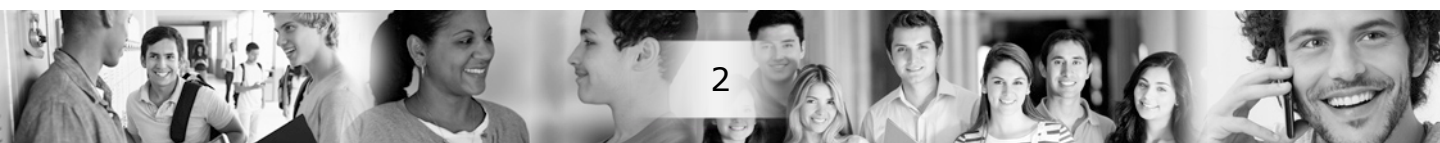
Let's Explore!

How to Use Code Coverage Metrics



Instructions: Read the text below.

- Monica:** I'm going to be writing the code for a new application. I'm still figuring out how coding works, and I want to clear up some doubts first. I'm aware that many developers love using the code coverage metric, but I think it's overrated.
- Jonathan:** It's good to be critical with the metrics you're using. From what I can tell you, once we have tested the code, the code coverage is going to let you know where and how much of the code remains untested. You're then going to be deciding whether it's worth testing the remaining code. This depends on how your application works. Likewise, if you're writing getters and setters, you're not going to be having to test them nor the code because that'll be generating automatically.
- Monica:** So, by the time I prove that the application is working fine, I'll have finished everything, and I won't be needing any further testing, right?
- Jonathan:** Most likely not. You'll be needing more tests if there's an important malfunction in the application. On the other hand, we cannot trust the code coverage metric when it comes to the assessment of the quality of the tested code. The metric will only tell you how much of the code you'll have tested but not how good the testing was. Therefore, you'll be exercising caution when interpreting the results of the code coverage metric.
- Monica:** Alright, this metric won't be assessing the quality of tests, but showing which lines of the code were untested. However, as I'm following a test-driven development method, I'm supposed to have a high percentage of covered code unlike other methods where testing is not at the core of the development process, am I right?
- Jonathan:** Exactly, by the time you get to 80% of code coverage, you'll have tested the necessary.





Let's Practice!

How important is to use code coverage metrics?

Instructions: Select the correct answer.

1. What will code coverage allow users to do?
 - a. It will be giving you access for you to know where and how much of the code remained untested.
 - b. It will assess the quality of tests and show which lines of the code were untested.
 - c. It will be giving programmers ideas about the need to test the remaining code.
 - d. It won't tell you how much of the code you'll have tested but how good the testing was.
 - e. Code coverage metric is overrated.



Let's Practice!

Code coverage aspects you need to know.

Instructions: Read the sentences. Choose the correct answer.

1. You _____ 80% of code coverage by the time your code has been tested.
Hint: Take into account the auxiliary verb.
a. will have getting b. will have gotten c. won't have d. will having
2. By the time you finish, you _____ that the application is working correctly. You shouldn't need any further testing.
a. will have prove b. will have proving c. will have proved d. will proving
3. Once I get a high percentage of code coverage, I _____ the good state of a program.
Hint: Take into account the auxiliary verb.
a. will have guaranteed b. will have guaranteeing c. will be guaranteed d. will guarantying
4. Without an effective code coverage you _____ aware of the performance of an application.
Hint: Take into account the auxiliary verb.
a. won't having be b. won't have been c. won't have being d. won't being
5. By being critical with the metrics to use, you _____ mistakes.
a. will have preventing b. will be preventing c. will be prevented d. will preventing
6. I'll _____ the number of tests if the application shows inconsistencies.
a. be increasing b. have increasing c. be increased d. increased



Extension Activity

Expressing opinions

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Think about the experiences you've had with code coverage testing.
- How important do you consider it is for your work? Discuss it with another partner.
- Write a 300 words blog entry in which you express your position about it, talk about your experience and give some tips to other programmers. Use future perfect and continuous.
- Share your opinions with your partners.

Self-Assessment

Yes No

- I share some experiences I've had with code coverage testing. ☐ ☐
- I discuss with my partners about the importance of code covering. ☐ ☐
- I Write a 300 words blog entry in which I express your position about it ☐ ☐
- I share my opinions with your partners ☐ ☐



Target Words

- getter** (noun): a method that gets the value of a variable in the process of limiting the possible changes that can be applied on a code.
e.g. If you're writing getters and setters, you're not going to be having to test them nor the code that'll be generating automatically.
- setter** (noun): a method that sets the value of a variable in the process of limiting the possible changes that can be applied on a code.
e.g. If you're writing getters and setters, you're not going to be having to test them nor the code that'll be generating automatically.
- exercise caution** (expression): to be careful and critical with something.
e.g. You'll be exercising caution when interpreting the results of the code coverage metric.
- overrated** (adjective): something that is valued more than it should.
e.g. I'm aware that many developers love using code coverage metric, but I think it's overrated.
- malfunction** (noun): to fail to work properly.
e.g. You'll be needing more tests if there's and important malfunction in the application.
- test-driven development** (noun): it is software used to develop and test the code coverage.
e.g. As I'm following a test-driven development method, I'm supposed to have a high percentage of covered code unlike other methods where testing is not at the core of the development process.



Learning Points

Future continuous

We use the future continuous to talk about actions that will be happening during a period of time in the future.

We can use the **future continuous** in different situations:

1. When projecting ourselves into the future.

Example:

*This afternoon, we **will be evaluating** our metrics.*

2. When predicting something that can happen in the future.

Example:

*Next month, our manager **is going to be evaluating** the performance of the project.*

3. When talking about continuous events that will take place in the future.

Example:

*When we test the code, we **will be evaluating** the code coverage metric.*

The structure of the future continuous is as follows:

Type of sentence	Structure	Example
Affirmative	Subject + will/be going to + be + verb in -ing	<i>The team will be using code coverage after testing.</i> <i>The team is going to be using code coverage after testing.</i>
Negative	Subject + will not/be not going to + be + verb in -ing	<i>The team will not be using code coverage after testing.</i> <i>The team is not going to be using code coverage after testing.</i>
Interrogative	* Will + subject + be + verb in -ing * Verb to be + subject + going to be + verb in -ing	<i>Will the team be using code coverage after testing?</i> <i>Is the team going to be using code coverage after testing?</i>
Interrogative with 'wh' word	* 'Wh' word + will + subject + be + verb in -ing * 'Wh' word + verb to be + subject + going to be + verb in -ing	<i>How will the team be using code coverage after testing?</i> <i>When is the team going to be using code coverage after testing?</i>

Examples

- *I'm going to be writing the code for a new application.*
- *Once we have tested the code, code coverage is going to be letting you know where and how much of the code remained untested.*
- *I won't be needing any further testing, right?*
- *You'll be needing more tests if there's an important malfunction in the application.*



Learning Points

Future perfect

We use the future perfect to refer to an action that will have finished at a point in the future.

Although this tense is not too common, we can use it when we want to talk about an action that will take place in the future but that will be finished before the event we are mentioning takes place.

Example: ***By next month, we will have finished the project.***

We can also use this tense when we want to clarify that an action in the future occurs before another action.

Example: ***Before we use code coverage as a metric, we will have determined what testing parameters to use.***

Take a look at the previous examples. Notice that when we have the future perfect tense, we specify the moment or event that takes place after the action of the future perfect. This moment or event is usually introduced by the preposition 'by' or by the action that will take place afterwards.

Type of sentence	Structure	Example
Affirmative	Subject + will have + past participle of the main verb	<i>We will have tested the code by tomorrow.</i>
Negative	Subject + will not have + past participle of main verb	<i>We will not have tested the code by tomorrow.</i>
Interrogative	Will + subject + have + past participle of the main verb.	<i>Will we have tested the code by tomorrow?</i>

Examples

- This metric will only tell you how much of the code you'll have tested but not how good the testing was.*
- By the time you get to 80% of code coverage, you'll have tested the necessary.*
- By the time I prove that the application is working fine, I'll have finished everything, and I won't be needing any further testing, right?*